

The Binding Number of a Graph and its Cliques

Jeremy Lyle^{*,a,1}, Wayne Goddard^b

^a*Department of Mathematics, The University of Southern Mississippi, Hattiesburg, MS USA*

^b*School of Computing and Department of Mathematical Sciences, Clemson University, Clemson, SC USA*

Abstract

We consider the binding numbers of K_r -free graphs, and improve upper bounds on the binding number which force a graph to contain a clique of order r . For the case $r = 4$, we provide a construction for K_4 -free graphs which have a larger binding number than previously known constructions. This leads to a counterexample to a conjecture by Caro regarding the neighborhoods of independent sets.

Key words: binding number, K_r -free graphs

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1. Introduction

Woodall [11] defined the binding number of a graph as follows. If $S \subseteq V(G)$, then we write the open neighborhood of the set S as $N(S) = \bigcup_{v \in S} N(v)$. The binding number of G , denoted $\text{bind}(G)$, is given by

$$\text{bind}(G) = \min_{S \in \mathcal{S}(G)} \frac{|N(S)|}{|S|}, \text{ where } \mathcal{S}(G) = \{S \subseteq V(G) : S \neq \emptyset, N(S) \neq V(G)\}.$$

In [11], Woodall showed that a binding number of at least $3/2$ implies that the graph is hamiltonian. Additionally, Shi Rong-Hua showed in [9] that a binding number of $3/2$ also implies that the graph contains a triangle, and furthermore, the graph is pancyclic [10] (contains a cycle of every intermediate length). (See [4] for a short proof for a triangle.) The binding number $3/2$ is best possible for the existence of a triangle (see Subsection 2.1).

Kane and Mohanty [6] observed the following theorem, which follows from a result of Andrásfai, Erdős, and Sós [1]:

Theorem 1 ([6]). *For any K_r -free graph G , $\text{bind}(G) \leq r - 4/3$.*

In addition, they observed that there are K_r -free graphs with binding number $r - 2$, such as the tensor product $K_{r-1} \otimes K_m$ where $m \geq (r - 2)^2 + 1$.

In Subsection 2.2 we will improve on this result when $r = 4$ by showing that there are K_4 -free graphs with binding number arbitrarily close to $\frac{1}{4}(3 + \sqrt{33}) \approx 2.186$. In Subsection 2.3 we will use a similar construction to disprove a conjecture of Caro. In Section 3 we will improve on Theorem 1 by showing that a K_4 -free graph has binding number at most $\frac{1}{6}(5 + \sqrt{91}) \approx 2.423$, and if $r \geq 5$ then a K_r -free graph has binding number at most $r - 3/2 - \frac{1}{4r-6}$.

^{*}Corresponding Author

¹Research performed while at Department of Mathematical Sciences, Clemson University

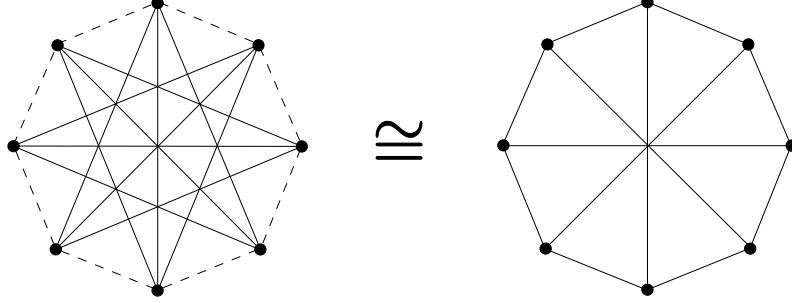


Figure 1: The graph Γ_3 , isomorphic to the Möbius ladder on 8 vertices.

2. K_r -free Graphs with Large Binding Number

2.1. Triangle-free graphs

For $i \geq 2$, let $\Gamma_i = \overline{C_{3i-1}^{i-1}}$, the complement of the $(i-1)$ -th power of the cycle C_{3i-1} . For example, Γ_2 is isomorphic to the 5-cycle and Γ_3 is isomorphic to the Möbius ladder on 8 vertices (See Figure 1).

This family is well known. We state the following lemma to summarize some known properties of these graphs.

Lemma 1. *The graph Γ_i is a 3-colorable, i -regular, triangle-free graph, and has the property that any set of b vertices has an open neighborhood of at least $\min\{b+i-1, n\}$ vertices, where $n = |V(\Gamma_i)| = 3i-1$.*

PROOF. It is well-known [11, 7] that the graph Γ_i is 3-colorable, i -regular, and contains no triangles. To show the property about the open neighborhoods of sets, we use induction on b . It is true if $b = 1$, as Γ_i is i -regular. Assume for induction that any set of $b-1$ vertices has an open neighborhood of order at least $\min\{b+i-1, n\}$. Let S be a set of order b such that $N(S) \neq V(\Gamma_i)$, and choose $v \in V(\Gamma_i) \setminus N(S)$. If the vertices of Γ_i are spaced unit distance apart around a circle in the same order as in the original cycle C_{3i-1} , then all vertices of S lie within the closed arc of length $2(i-1)$ centered on v . Let u be the vertex of S furthest from v clockwise within this arc. Let w be the first vertex of $N(u)$ encountered when travelling counterclockwise from v . Then w cannot be adjacent to any other vertex of S . Thus $|N(S)| \geq |N(S \setminus \{u\})| + 1 \geq ((b-1) + i - 1) + 1 = b - i + 1$, by the induction hypothesis. ■

In light of these properties, it is not hard to show that the binding number of this family of graphs tends to $3/2$ from below, as noted by Woodall [11]. This family figures in other results. For example, Chen, Jin, and Koh [3] showed that a 3-colorable triangle-free graph with minimum degree $\delta(G) > n/3$ must be homomorphic to some Γ_i . (The definition of their family looks different but it is the same family of graphs.)

2.2. K_4 -free graphs

We improve on the lower bound of Kane and Mohanty by constructing K_4 -free graphs with binding number larger than 2. Let $\Gamma_i = \overline{C_{3i-1}^{i-1}}$ as before. Define the graph $G(i, a) = \Gamma_i \vee \overline{K_a}$, the join of the triangle-free graph Γ_i and an independent set of a vertices. This graph is K_4 -free. It is not hard to show:

Lemma 2. *For $i \geq 2$ and $a \geq 1$, the binding number of $G = G(i, a)$ is given by*

$$\text{bind}(G) = \min \left\{ \frac{3i-2+a}{2i-1}, \frac{3i-1}{a} \right\}.$$

PROOF. For a set S with $N(S) \neq V(G)$, we can take vertices either only from Γ_i or only from $\overline{K_a}$.

By Lemma 1, if S is any set of b vertices of Γ_i then $|N_{\Gamma_i}(S)| \geq b + i - 1$, with equality if the b vertices are consecutive and $b \leq 2i$. To ensure that $N_G(S) \neq V(G)$, we require $b \leq 2i - 1$, so that $b + i - 1 < |V(\Gamma_i)| = 3i - 1$. Recall that $\mathcal{S}(G) = \{S \subseteq V(G) : S \neq \emptyset, N(S) \neq V(G)\}$. Thus

$$\min \{|N_G(S)|/|S| : S \in \mathcal{S}(G), S \subseteq V(\Gamma_i)\} = \min_b \frac{b + i - 1 + a}{b} = \frac{3i - 2 + a}{2i - 1},$$

since the middle expression is decreasing in b , and therefore achieves its minimum when $b = 2i - 1$. On the other hand,

$$\min \{|N_G(S)|/|S| : S \in \mathcal{S}(G), S \subseteq V(\overline{K_a})\} = \frac{3i - 1}{a},$$

since the minimum is taken when $S = V(\overline{K_a})$. Thus $\text{bind}(G)$ is the minimum of the above two expressions. ■

Since one expression in the formula of Lemma 2 is increasing in a and the other expression decreasing in a , we let $a(i)$ be the nearest integer to the value of a where the two expressions are equal. It can be calculated that

$$\lim_{i \rightarrow \infty} \text{bind}(G(i, a(i))) = \frac{3 + \sqrt{33}}{4} \approx 2.186.$$

Theorem 2. *There are K_4 -free graphs with binding number arbitrarily close to $\frac{1}{4}(3 + \sqrt{33}) \approx 2.186$.*

2.3. A counterexample to a conjecture of Caro

An extension of this family of graphs disproves the following conjecture, which was made by Yair Caro in a postscript to [8]:

Conjecture 1 (Caro). *Let $r \geq 2$ be an integer and let G be a graph of order n such that $|N(X)| > (r - 2)(n + |X|)/r$ for every independent subset X of $V(G)$. Then G contains a copy of K_r .*

For $r = 2$, the conjecture simply states that G contains an edge if $|N(X)| > 0$ for every independent subset X , and the main result of [8] showed the conjecture true for $r = 3$. However, the conjecture is false for $r \geq 4$, and the graphs can be constructed in the following way. Let $\Gamma_i = \overline{C_{3i-1}^{i-1}}$, as before. Now, define the graph $G(i, r, a) = \Gamma_i \vee (\vee_{j=1}^{r-3} \overline{K_a})$, the join of the triangle-free graph Γ_i and $r - 3$ independent sets of a vertices, which has order $n = 3i - 1 + (r - 3)a$. It is immediate that $G(i, r, a)$ is K_r -free.

Consider $G(i, r, a)$ with $r \geq 4$. Since any independent set in a join of subgraphs is contained in only one of the subgraphs, there are two cases to consider. First, an independent set X contained in a subgraph of the form $\overline{K_a}$ has $|N(X)| = n - a$ and $|X| \leq a$. It follows by some algebra that the hypothesis of the conjecture holds for all such X if and only if

$$2a < 3i - 1.$$

Second, an independent set X contained in the subgraph Γ_i has $|N(X)| \geq |X| + i - 1 + (r - 3)a$ (with equality if the vertices of X are consecutive), and $|X| \leq i$. It follows by some algebra that the hypothesis of the conjecture holds for all such X ($|X| = 1$ is the hardest) if and only if

$$a > i.$$

If $i < a < \frac{1}{2}(3i - 1)$, then we have a K_r -free graph which satisfies the hypothesis of the conjecture. Thus, for example, $G(4, r, 5)$ is a counterexample.

Theorem 3. *Conjecture 1 is false for all $r \geq 4$.*

The graph $G(4, 4, 5)$ can be drawn as in Figure 2, where each of the eleven outside vertices is adjacent to all five vertices of $\overline{K_5}$.

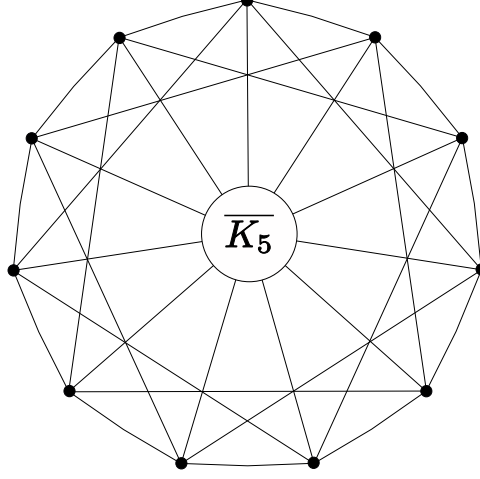


Figure 2: A counterexample to Caro's conjecture, $G(4, 4, 5)$

3. Upper Bounds on the Binding Number

Next, we consider finding upper bounds on the binding number of a K_r -free graph; first for $r = 4$, and then for $r > 4$. In both cases, we accomplish this by finding bounds on the independence number of a K_r -free graph with large minimum degree.

3.1. Upper bounds on the binding number of K_4 -free graphs

For K_4 -free graphs, we begin by proving an easy strengthening of a result in [1] for degree-sums:

Lemma 3. *Let G be a triangle-free graph on n vertices containing a 5-cycle C . Then there exists two consecutive vertices a and b on C such that $d(a) + d(b) \leq 4n/5$.*

PROOF. Since each vertex in G is adjacent to at most two vertices of C , $\sum_{v \in V(C)} d(v) \leq 2n$. Thus the average sum of the degrees of two consecutive vertices of C is at most $4n/5$, as required. ■

We also need the result of Brandt [2] that all maximal K_4 -free graphs with $\delta(G) > 4n/7$ that are not 3-colorable contain the 5-wheel as a subgraph.

Theorem 4. *Let G be a K_4 -free graph with $\delta(G) > 4n/7$. Then the independence number of G is at least*

$$\min \left\{ \frac{n}{3}, \left(\frac{11}{5} \right) \delta(G) - n \right\}.$$

PROOF. We may assume that G is maximal K_4 -free. If G is 3-colorable, then it has an independent set of at least $n/3$ vertices. Otherwise, by the minimum degree condition, we know from Brandt's result that G contains a 5-wheel. Let u be the center of the wheel, so that $N(u) = N_G(u)$ contains a 5-cycle C . Let H be the subgraph of G induced by $N_G(u)$; then H is triangle-free. By Lemma 3, C contains consecutive vertices a, b such that $d_H(a) + d_H(b) \leq 4|V(H)|/5 = 4|N_G(u)|/5$. Let $T = N_G(a) \cap N_G(b)$, which is an independent set contained in $V(G) \setminus V(H)$. Then,

$$\begin{aligned} |T| &= |N_G(a) \setminus N_H(a)| + |N_G(b) \setminus N_H(b)| - |V(G) \setminus V(H)| \\ &\geq 2\delta(G) - 4|N_G(u)|/5 - (n - |N_G(u)|) \\ &= 2\delta(G) - n + |N_G(u)|/5 \geq \frac{11}{5}\delta(G) - n. \quad \blacksquare \end{aligned}$$

This provides a simple upper bound on the binding number of all K_4 -free graphs.

Theorem 5. *For any K_4 -free graph G ,*

$$\text{bind}(G) \leq \frac{1}{6}(5 + \sqrt{91}) \approx 2.423.$$

PROOF. If i is the order of the largest independent set, as before we have $\text{bind}(G) \leq (n - i)/i = n/i - 1$. By Theorem 4, $i \geq \min\{n/3, (11/5)\delta(G) - n\}$. If $i \geq \frac{n}{3}$, then we have $\text{bind}(G) \leq 2$. So assume that $i \geq (11/5)\delta(G) - n$. Then we get $\text{bind}(G) \leq (10n - 11\delta(G))/(11\delta(G) - 5n)$. This decreases with $\delta(G)$. Also, we can choose the nonneighborhood of a minimum-degree vertex, and get $\text{bind}(G) \leq (n - 1)/(n - \delta(G)) < n/(n - \delta(G))$. This increases with $\delta(G)$. Then, at $\delta(G) = ((16 - \sqrt{91})/11)n$, these two values are equal, and give a maximum binding number of $\frac{1}{6}(5 + \sqrt{91}) \approx 2.423$. ■

We conjecture that the lower bound (2.186...) is the correct threshold.

3.2. Upper bounds on the binding number of K_r -free graphs, $r \geq 5$

Large independent sets can also be found in K_r -free graphs ($r \geq 5$) with large minimum degree. We use a result of [5, 7], that if G is K_r -free ($r \geq 5$) and $\delta(G) > (2r^2 - 8r + 7)n/(2r^2 - 6r + 4)$, then there exists an independent set of at least $n/(r - 1)$ vertices.

Theorem 6. *For $r \geq 5$ and any K_r -free graph G ,*

$$\text{bind}(G) < r - 3/2 - \frac{1}{4r - 6}.$$

PROOF. If $\delta(G) \leq (2r^2 - 8r + 7)n/(2r^2 - 6r + 4)$, then taking S to be the set of vertices nonadjacent to a vertex of minimum degree, we get

$$\text{bind}(G) < \frac{n}{n - \delta(G)} \leq \frac{1}{1 - \frac{2r^2 - 8r + 7}{2r^2 - 6r + 4}} = r - \frac{3}{2} - \frac{1}{4r - 6}.$$

Otherwise, we can apply the result of [5] to find an independent set of at least $i = n/(r - 1)$ vertices, so that

$$\text{bind}(G) < \frac{n - i}{i} = \frac{n}{i} - 1 = (r - 1) - 1 = r - 2. \quad \blacksquare$$

This improves on the upper bound $r - 4/3$ of Kane and Mohanty.

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